More importantly, we see that inspecting the locations in the middle, either the fiftieth or the fifty first numbers in the list, gives us the maximal amount of information. This is because once we do, we will be guaranteed to rule out *half* of the list, regardless of whether our target number is greater or less than the number in the middle. This is not the case for other locations. As we saw earlier, we may be able to rule out ninety numbers in the list when looking at the tenth number, but this only happens with 0.1 probability. When looking at the first number, there's 99 percent probability that we will only be able to rule out one number.

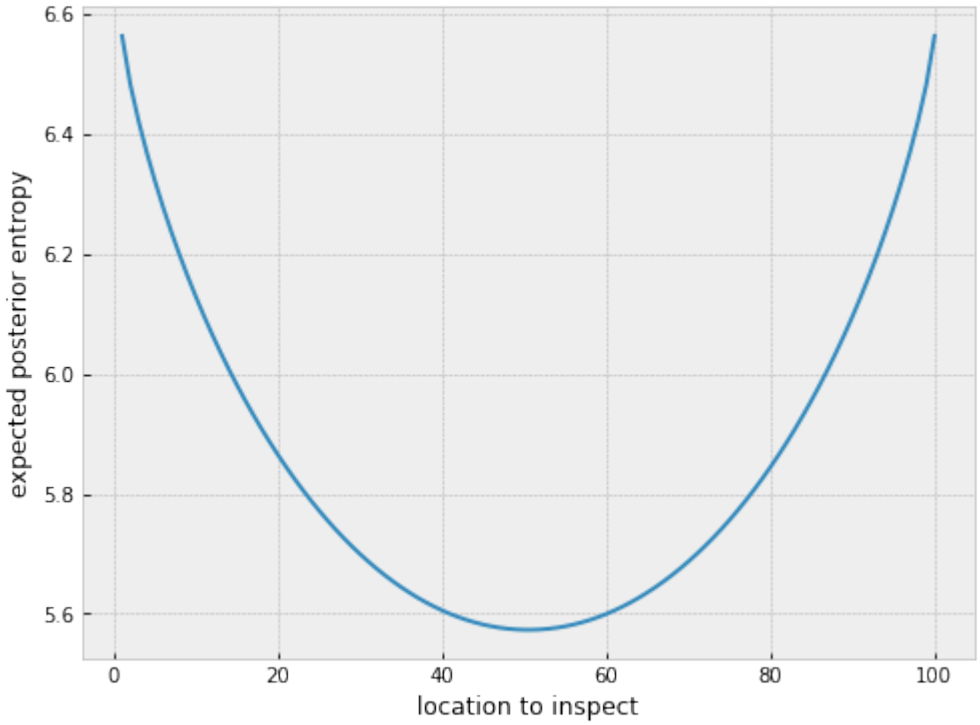


Figure 6.9 Expected posterior entropy in the location of the target as a function of the location within the list to inspect. The middle location is optimal, minimizing the expected entropy.

NOTE **All in all, inspecting the numbers in the middle maximizes the amount of information we gain on average.**

The rest of the search for the target follows the same procedure: computing the expected posterior entropy that will result from each decision and then choosing the decision that minimizes that entropy. As the probability distribution we work with is always a uniform distribution after each update, the optimal number to inspect is always the one in the middle of the array that hasn't been ruled out.

This is exactly the strategy of binary search! From the perspective of information theory, binary search is the optimal solution to the problem of search for a number in a sorted list.

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